NSAT Phase 2 - Question Paper 2

Sr. No.	Section Name		No. of Questions
1	Mathematics	10th Class Maths	15
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2	General Aptitude	Logic & Data Interpretation	20
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SECTION – 1: Advance Mathematics

1. Let A be a set containing five elements. If B and C are two subsets of A such that B $\subseteq$ C, then the number of ordered pairs (B, C) is

(1) 243 (2) 32 (3) 100 (4) 140 (5) 64

2. What is the number of different nine-letter words beginning and ending with a consonant which can be made out of the letters of the word "JUBILANCE"?

(1) 1,00,800 (2) 45,360 (3) 72,000 (4) 40,320 (5) None of these

3. The value of $\log_e\left[\lim\limits_{x\to\infty}\left(\frac{x+6}{x+1}\right)^{x+4}\right]$ is

(1) 2 (2) e^5 (3) 5 (4) 0 (5) Infinity

4. If the curve $y = a^2(1-x)x^2$ is bounded by the lines 8y = 1 and x = 1, then the area bounded by the curve is

(1) $\frac{15+5\sqrt{5}}{384}$ at a = 1 (2) $\frac{1}{4}$ at a = 0 (3) $\frac{1}{8}$ at a = 0

(4) $\frac{17+5\sqrt{5}}{2304}$ at a = 1 (5) None of these

5. Consider the given integral.

$I = \int_{2\alpha}^{2\beta} \frac{|z^3|}{z^3}\,dz$, where $\alpha < \beta$

Among the following options, which one does not represent the value of the integral I?

(1) 2(β - α), if α < 0 < β (2) 2(α - β), if α < β < 0

(3) 2(β - α), if 0 < α < β (4) 2(α + β), if α < 0 < β

(5) both (1) and (3)

6. In a group of 100 people, it was found that 50 people like to travel by train, 40 like to travel by bus and 20 like to travel by car (there may be some people who do not like to travel by any of these three). If x and y be the maximum and minimum numbers of people who like to travel by all the three modes of transport, then the difference between x and y is

(1) 20 (2) 30 (3) 40 (4) 50 (5) Can't be determined

7. If A = {x : x I, x = 4n2 – 5n + 5, n [– 2, 2]}, then n(A) is equal to

(1) 28 (2) 29 (3) 30 (4) 5 (5) 27

8. Let $\vec{a} = 3\hat{i} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} - 4\hat{j} + 7\hat{k}$. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$. The magnitude $|x + y + z|$ is equal to

(1)12/7 (2) 33/7 (3) 22/7 (4) 88/7 (5) None of these

9. The number of solutions of the inequality $2^{\frac{1}{\sin^2\alpha_2}} \times 3^{\frac{1}{\sin^2\alpha_3}} \times 4^{\frac{1}{\sin^2\alpha_4}} \times 5^{\frac{1}{\sin^2\alpha_5}} \leq 120$, where $\alpha_i \in (0, 2\pi)$ for i = 2, 3, 4, 5, is

(1) 81 (2) 9 (3) 16 (4) 8 (5) 5

10. Consider the equation $(x + 1)y' = x^2$. Find the value of y.

(1) $\frac{x^2}{2} - x + \log|x + 1| + c$ (2) $\frac{x^2}{2} - x + \log|x - 1| + c$ (3) $+ x + \log|x + 1| + c$

(4) $- x + \log|x + 1| + c$ (5) None of these

11. If a function $y = \frac{(4a+3x)^{1/2} - (x+6a)^{1/2}}{(2a+3x)^{1/2} - (3a+4x)^{1/2}}$ and $m = \lim_{x\to\infty} y$, find the value of $-2m/(1 + \sqrt{3})$.

(1) 2 (2) $(i)^2\log_2 4$ (3) $(i)^4\log_3 4$ (4) -2 (5) None of these

12. 150 students are taking at least one of the courses out of Chemical Engineering and Petroleum Engineering. How many of the 150 students are taking Chemical Engineering, but not Petroleum Engineering?

Statements:

I. 48 students are taking Chemical Engineering and Petroleum Engineering.

II. The number of students taking Petroleum Engineering, but not Chemical Engineering is the same as the number of students taking Chemical Engineering, but not Petroleum Engineering.

III. 32% students are taking Chemical Engineering and Petroleum Engineering.

(1) III only (2) I only (3) Either I or II (4) Either I or III (5) None of these

13. Consider a function, f(x), which is defined differently depending on whether x is even or odd. When x is an even number, f(x) = g(x) [g(x) + 1], and when x is an odd number then f(x) =g(x).

Also, function $g(x) = \begin{cases} x & \text{when x is even} \\ x+3 & \text{when x is odd} \end{cases}$

For which integral value of x, the equation 8f(x + 1) - f(x) = 2 is satisfied?

(1) -2 (2) 2 (3) 10 (4) -10 (5) Can't find

14. Determine the value of r, for which the f(x) = $\sum_{x=1}^{x=r}\left[\frac{1}{6}+\frac{x}{36}\right]$ is equal to 108, where [m] denotes largest integer less than or equal to m.

(1) 98 (2) 101 (3) 103 (4) 107 (5) None of these

15. If $3+\sqrt{2}$ and $2-\sqrt{7}$ respectively are the roots of the quadratic equation $mx^2 + nx + p = 0$ and $ay^2 + by + c = 0$, find the value of $(m - a)^2 + (n - 1.5b)^2 + (3p + 7c)^2$.

(1) 0 (2) 1 (3) 4 (4) 9 (5) 14

SECTION – 2: Basic Mathematics

16. In a bookstore, the average price of books sold increased by Rs. 10 when a new collection of books were added. The average price of the new collection of books is Rs. 80 more than half the average price of the original books. If the ratio of the number of original books to the number of books from the new collection is 3 : 2, what is the average price of the new collection of books?

(1) 55 (2) 80 (3) 135 (4) 155 (5) None of these

17. In a company, the total number of employees increased by 20% from the last year to the current year.
If the number of male employees increased by 15% and the number of female employees increased by 25%, what is the percentage increase/decrease in the ratio of male to female employees?

(1) 8% decrease (2) 8% increase (3) 16% increase (4) 16% decrease (5) None of these

18. In a school, the ratio of the number of boys to that of girls is 8 : 5. The total number of students is 91. The average height of the boys is 160 cm, while the average height of the girls is 150 cm. When the heights of the girls are arranged in ascending order, they form an arithmetic progression (AP) and the height of the third girl from the last is 165 cm. Find the height of the tallest girl.

(1) 165 cm (2) 167 cm (3) 169 cm (4) 171 cm (5) None of these

19. All the sides of a right-angled triangle have integral values. If one of the sides is 40 metres, what can be the value of the hypotenuse, in metres, of this triangle?

(1) 299 (2) 301 (3) 399 (4) 401 (5) None of these

20. Let a, b and c be positive real numbers satisfying the following conditions:
$a^3 = b^5 = c^7$
$(a^2)(b^3)(c^4) = 2^{(12)}$
Find the value of $(a^6)(b^4)(c^3)$.

(1) $2^{\frac{1,017}{193}}$ (2) $2^{\frac{1,317}{162}}$ (3) $2^{\frac{4,068}{193}}$ (4) $2^{\frac{4,068}{165}}$ (5) None of these

21. Anil and Bobby are playing a number game. Anil thinks of a secret positive integer between 1 and 100, inclusive. He doubles the number and tells the result to Bobby. Bobby, in turn, adds 20 to the number and shares it back with Anil. They keep doing this, alternating between doubling and adding 20, until the number exceeds 200. The first person whose result exceeds 200 wins the game. Let 'x' be the smallest initial number that results in a win for Anil. What is the value of digit 'x'?

(1) 8 (2) 6 (3) 4 (4) 2 (5) None of these

22. A tug of war competition is organised to test the power of a new Cybertruck against a military tank. On the day of competition it starts raining but preparation continues. The cybertruck and the tank are placed in opposite directions on an elevated platform 5 inches above the ground. This platform is made of wooden planks and has small gaps in between to give good friction to vehicles. Rear ends of both the vehicles are marked as point T1 & T2 between which a 100 inch wide normal rope is tied. Point T1 & T2 are 55 inch & 85 inch above the platform. Once rain stops the competition starts.

At one point of time in the competition, a drop of water freely falls from the lowest point of rope, freely passes through the gap between planks and reaches ground in 5/root(193) seconds. At this time, what is the distance between point T1 & T2.

Supplementary information:
h = ut + (gt^2)/2 where u is the initial speed of an object falling from height h, taking t time to reach ground. Assume g = 386 inches/sec^2 and no air resistance is present.

(1) 100 Inches (2) 75 Inches (3) 50 Inches (4) 25 Inches (5) None of these

23. Let a and b be real numbers. If $a^2 - ab - 3a = 10$ and $b^2 - ab + 3b = 18$, then what is the value of (a - 2b)?

(1) -9/2 (2) 5/2 (3) 7/2 (4) 23/2 (5) None of these

24. Assume σ is a prime number and σ^7 is the smallest common multiple of α and **β**. How many unique smallest common multiples are possible for α^4 and **β**^8?

(1) Infinite (2) 1 (3) 5 (4) Can't be determined (5) None of these

25. In a dynamic business, Alex encounters fluctuating profit margins in his sales efforts. Initially, he sells a camera at a 12% loss and a tablet at an 18% gain, resulting in a profit of Rs. 1,800. However, after changing his approach, he sells the camera at a 7% gain and the tablet at a 21% gain, which yields a profit of Rs. 2,800. What was the original cost of the tablet, in rupees?

(1) 14,322.44 (2) 11,232.22 (3) 12,222.22 (4) 15,332.44 (5) None of these

26. A cylindrical water tank has a diameter of 2 metres and a height of 'h' metres. The tank is initially empty. A tap is opened, and water flows into the tank at a constant rate of $\frac{1}{12}$ litres per second. Simultaneously, there is a hole at the bottom of the tank through which water leaks out at a constant rate of $\frac{1}{15}$ litres per second. The tank is completely filled with water in 77 hours. What is the value of 'h'?

(1) 1.20 (2) 1.47 (3) 1.64 (4) 2.67 (5) None of these

27. Consider three communication towers labelled as X, Y, and Z standing in a particular area. Tower X has a height of 80 metres, and tower Y has a height of 120 metres. These towers are constructed to be perfectly vertical. Now, a cable is attached from the top of tower X to the base of tower Y, and another cable is attached from the top of tower Y to the base of tower X. The point where these cables cross each other is referred to as point Z', which coincides with the highest point of tower Z. What is the height (in metres) of tower Z?

(1) 62 (2) 64 (3) 58 (4) 48 (5) None of these

28. How many factors of $3^2 \times 6^6 \times 9^3 \times 12^5$ are perfect cubes greater than 1?

(1) 31 (2) 38 (3) 41 (4) 43 (5) None of these

29. Consider two medical centres, Centre X and Centre Y. Centre X admitted a certain number of patients with a particular infectious disease, and it was 21 patients less than the number of patients admitted to Centre Y with the same disease. All patients in both medical centres eventually recovered. The total sum of recovery days for patients in Centre X was 200 days, while for patients in Centre Y, it was 2 days more than the 75% of the number of recovery days for patients in Centre X. If the average number of recovery days for patients admitted to Centre X was 3 days more than the average number of recovery days for patients admitted to Centre Y, then how many patients were admitted to Centre Y?

(1) 35 (2) 25 (3) 46 (4) 56 (5) None of these

30. In a lovely neighbourhood, three enthusiastic gardeners named Lila, Ryansh, and Eshan are working on a gardening project. Each gardener has their own unique expertise and can complete the gardening task individually in 90, 120, and 150 days, respectively. Excited to create a beautiful garden together, they decide to start the project as a team.
However, as the days go by, Lila receives an urgent call and has to leave after 20 days of working together. Additionally, with only 8 days left until the completion of the gardening project, Ryansh faces a sudden family commitment and also has to leave. How many days will it take to complete the entire gardening project from the very beginning to the final completion?

(1) $56\frac{4}{27}$ (2) $56\frac{8}{27}$ (3) $56\frac{4}{7}$ (4) $56\frac{5}{9}$ (5) None of these

SECTION – 3: GENERAL APTITUDE

Directions for questions 31 – 32: Read the following information carefully and answer the questions that follow.

In a family, P, Q, R, S, T, U, V and W are the eight members, out of which there are three couples and as many generations. Each of the family members belongs to a different profession - Scientist, Writer, Artist, Musician, Teacher, Engineer, Colonel and Lawyer, not necessarily in the same order. Each couple has at least one child.

1. S and T are the children of P, the Writer.
2. The Teacher is married to the Engineer and the Artist is a female.
3. Q is the Musician and is not yet married, and U, the Colonel, is her cousin.
4. W is younger than V and they are of different genders.
5. The Teacher and R are sisters-in law.
6. T has a male child and is married to the Lawyer.

31. How is P related to W?

(1) Mother-in-law (2) Father-in-law (3) Daughter-in-law
(4) Son-in-law (5) Father

32. How is the Musician related to the Scientist?

(1) Cousin (2) Daughter (3) Daughter-in-law
(4) Aunt (5) Niece

Directions for questions 33 – 34: Study the following information carefully and answer the questions.

In a park, there are two paths, XY and AB. The XY path represents the north and south directions, with X indicating north and Y indicating south. The AB path represents the west and east directions, with A indicating west and B indicating east. The point of intersection between these two paths is denoted as O, and the distances from O to various points are recorded as follows: XO = 20 m, OY = 19 m, OA = 18 m, and OB = 30 m.

To explore the park, three individuals named Lila, Manas, and Sonia start walking in separate directions, following specific instructions. Lila starts her walk from point B by moving 5 m west, then turning right and walking 7 m. She then proceeds southwards for 30 m before making a left turn and walking 5 m. Manas, on the other hand, begins his walk from point A. He moves 25 m north and then turns right, covering a distance of 35 m in the eastward direction. Sonia starts walking from point X, where she moves 5 m north. She then makes two consecutive left turns and walks 7 m each time.

33. In which direction is Manas's current position with respect to point X?

(1) Southeast (2) Southwest (3) Northeast (4) Northwest (5) None of these

34. What is the shortest distance between the current positions of Sonia and Manas?

(1) 14 m (2) 24 m (3) 25 m (4) 15 m (5) None of these

Directions: Study the following information to answer the given question.

Message	Coding
they need my book often	kin kic kih kiy kif
they do not require book	kic kit kih kiv kib
do not need to read often	kif kis kit kib kin kil
my book do need it	kic kiy kin kib kip
we require to do it	kib kiv kis kir kip

35. What is the code of 'require'?

(1) kip (2) kib (3) kiv (4) kir (5) kis

Directions: In the question below is given a group of letters and digits followed by four combinations of digit/symbol codes numbered (1), (2), (3) and (4). You have to determine which of the combinations correctly represents the group of letters and digits based on the following coding system and the conditions that follow, and mark the number of that combination as your answer. If none of the four combinations is correct, your answer is (5), i.e. 'None of these'.

Letter/Digit	G	E	A	6	2	R	O	5	K	N	9	P	U
Digit/Symbol Code	8	0	7	%	4	2	Z	@	&	D	1	$	*

Conditions:

(i) If the first letter is a consonant and the last digit is a prime number, both are to be coded as #.

(ii) If the first letter is a vowel and the last digit is divisible by 3, both are to be coded as the code for the last digit.

(iii) If the first letter is a vowel and the last digit is an odd number, then the codes for the first and last elements are to be interchanged.

36. ARONP2K5

(1) 72ZD$4&@ (2) @2DZ$4&7 (3) 7Z2D$4&@ (4) @2ZD$4&7 (5) None of these

Directions for questions 37 – 38: Read the following information carefully and answer the questions that follow.

K, L, M, N, O, P, Q, R, S, T, U and V are sitting in two rows. Six of them are facing north and six are facing south. They are sitting in such a way that every person is facing a person sitting in the other row. (Note: Facing the same direction means that if one is facing north, then the other also faces north and vice versa. Facing opposite directions means that if one is facing north, then the other faces south and vice versa.)

M is facing south. V is sitting third to the right of Q. L, who is second to the right of P, is facing N. O is not sitting at any extreme end and is facing the opposite direction to S. R and Q are sitting at the extreme ends, facing opposite directions, and they are not facing each other. U is facing the person who is sitting third to the left of V. O and T are sitting together. P is sitting second to the right of R. S and K are facing each other.

37. How many persons are sitting to the right of N?

(1) One (2) Two (3) Three (4) Four (5) Five

38. Which of the following statements is true?

(1) M is facing V. (2) R and O are neighbours. (3) O is facing south.

(4) M is sitting at an extreme end. (5) Q is facing U.

<u>Directions for questions 39 – 40:</u> Study the following information and answer the questions that follow.

Eight ministers, namely, Amit, Bikram, Chandu, Devi, Ekram, Faizal, Ganesh and Haider are sitting in a row facing north, but not necessarily in the same order. They all work for different ministries, such as consumer affairs, health, agriculture, planning commission, petroleum, education, civil aviation, and finance.

Devi is sitting to the immediate right of the minister who works for agriculture. The minister who works for health is sitting third to the right of Ekram. Ekram and Amit do not work for civil aviation. Amit is not sitting at an extreme end of the row. Three ministers are sitting between the ministers who work for consumer affairs and civil aviation, one of whom is sitting at an extreme end of the row. Ekram is the immediate neighbour of Amit and the minister who works for petroleum. Devi is not sitting at the second or third position from the left end of the row. Three ministers are sitting between the ministers who work for planning commission and education. The one who works for planning commission is sitting at the extreme right end of the row. The one who works for finance is sitting fifth to the left of the minister who works for planning commission. Ganesh, who does not work for consumer affairs, is sitting third to the right of Chandu. Ministers Bikram and Faizal are sitting adjacent to each other. Haider is not sitting to the right of Devi. At least one minister is sitting between Faizal and Ganesh. Amit does not work for agriculture. The one who works for consumer affairs is not sitting at the fourth position from the right end of the row. The minister who works for petroleum is neither at the second position from the right end of the row nor at the fourth position from the left end of the row.

39. Which of the following ministries does Ekram work for?

(1) Petroleum (2) Finance (3) Health (4) Consumer affairs (5) None of these

40. How many ministers are sitting to the right of the minister who is to the immediate right of Ganesh?

(1) 1 (2) 2 (3) 3 (4) 4 (5) None of these

Directions for questions 41 – 42: Study the following information carefully and answer the questions.

A prominent technology company unveiled its diverse range of cutting-edge software products - A, B, C, D, E, F, G, and H - in one of the technology hubs of India. The products are introduced one after the other over a span of six months in the year 2022. The strategic order of product launches follows a carefully planned sequence based on the following conditions:

- The revolutionary product E takes the lead, making its debut before both G and A.
- Following E's successful launch, the spotlight shifts to product A, making its grand entrance after H.
- The innovative product D takes its place in the lineup before E but after B, showcasing its remarkable capabilities.
- With a seamless flow, the cutting-edge product C joins the showcase, captivating the audience after E's introduction.
- Building up the momentum, product B claims its moment in the limelight, being launched before the impressive F.

Amidst much anticipation and excitement, the company introduces its array of groundbreaking software products, each designed to redefine the technological landscape in the bustling tech hub.

41. Out of the following, which could be true?

(1) The second product launched by the company is G.

(2) The third product launched by the company is B.

(3) The fourth product launched by the company is A.

(4) The fifth product launched by the company is C.

(5) None of these

42. If A is the fifth product launched by the company, which of the following will not be true?

(1) H is the first product to be launched. (2) D is the second product to be launched.

(3) E is the third product to be launched. (4) F is the fourth product to be launched.

(5) None of these

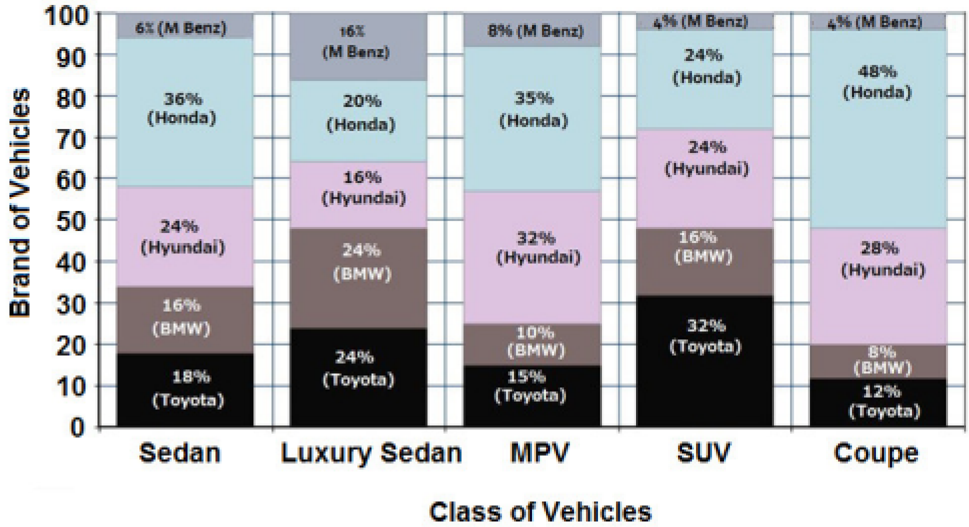
Directions for questions 43 – 44: Answer the questions on the basis of the following information.

The following bar graphs show the sales of 5 different brands of automakers in India across various classes of vehicles.

Bar Graph 1 gives the percentage breakup of the sales of different classes of vehicles across the different brands.

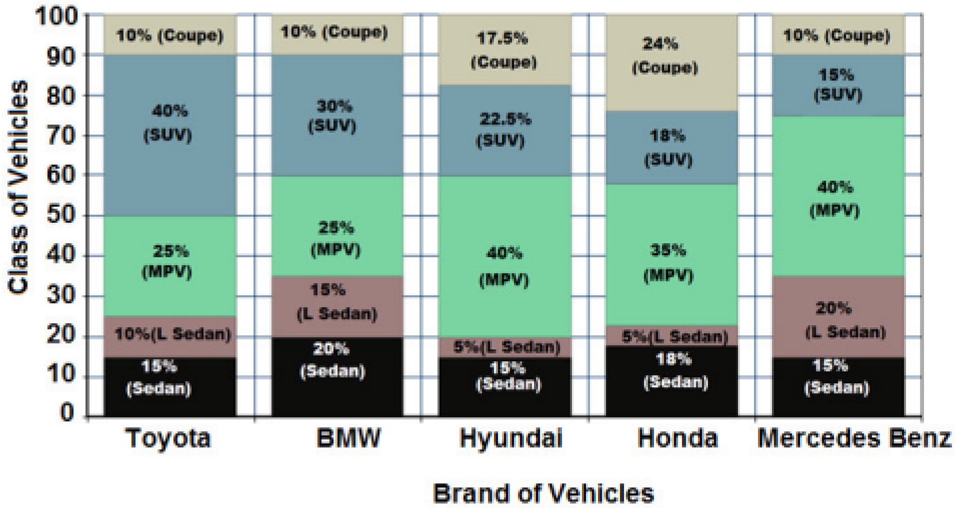
Bar Graph 2 gives the percentage breakup of the sales of vehicles of different brands across different classes of vehicles

Bar Graph 1



Note: M Benz is Mercedes Benz.

Bar Graph 2



Note: L Sedan is Luxury Sedan

43. What is the ratio of the number of vehicles sold across different classes in the order Sedan : Luxury Sedan : MPV : SUV : Coupe?

(1) 2 : 1 : 4 : 3 : 2 (2) 2 : 1 : 3 : 2 : 4 (3) 2 : 3 : 4 : 1 : 2

(4) 2 : 4 : 1 : 3 : 1 (5) None of these

44. What is the number of Luxury Sedans sold by Mercedes Benz?

(1)150 (2) 200 (3) 240 (4) Cannot be determined (5)None of these

Directions for questions 45 – 46: Study the following information carefully and answer the questions given below.

In a car race, different types of flags indicate different instructions for car drivers:

3 red flags = stop

2 red flags = turn left

1 red flag = turn right

3 green flags = go at 60 km/hr speed

2 green flags = go at 40 km/hr speed

1 green flag = go at 20 km/hr speed

A person starts driving from a point in the west direction and he encounters the following signals:

(i) Starting point – 1 green flag;

(ii) After 15 minutes, 1st signal – 2 red and 2 green flags;

(iii) After 18 minutes, 4th signal – 3 red flags;

(iv) After 24 minutes, 2nd signal – 1 red and 3 green flags;

(v) After 45 minutes, 3rd signal – 1 red and 2 green flags;

45. Find the total distance covered by the person up to the last signal.

(1) 74 km (2) 73 km (3) 76 km (4) 78 km (5) 75 km

46. What will be the driver's position with respect to the starting point if, instead of starting in the west direction, he starts in the south direction?

(1) 50 km to the south and 4 km to the west
(2) 50 km directly to the north-west
(3) 50 km to the north and 4 km to the west
(4) 50 km to the south and 4 km to the east
(5) None of these

Directions for questions 47 – 48: Study the following information and answer the questions.

Ludo Queen, Do Patti, and Zummy are the three most played online game genres in Asia. 320 million online gamers are interested only in Do Patti. Online gamers who are interested in both Do Patti and Ludo Queen together make up 125% of all online gamers who are only interested in Do Patti.

Online gamers who are interested in all three types of games make up 62.5% of the total number of online gamers interested in both Do Patti and Ludo Queen. 50% of the sum of the number of online gamers interested in only Do Patti and the number of gamers interested in all three types of online games is the number of online gamers interested in both Ludo Queen and Zummy but not Do Patti. The number of online gamers who are interested in only Zummy is 20% more than the number of online gamers who are interested in both Ludo Queen and Zummy but not in Do Patti. The number of online gamers interested in Ludo Queen is the same as the number of online gamers interested in Zummy. The number of online gamers interested in only Ludo Queen is half the number of online gamers interested in more than one type of online game.

47. What is the number of online gamers who are interested in only Zummy?

(1) 320 million (2) 601 million (3) 342 million (4) 285 million (5) None of these

48. What is the total number of online gamers who are interested only in one online game?

(1) 1,286 million (2) 1,304 million (3) 1,339 million (4) 1,155 million (5) None of these

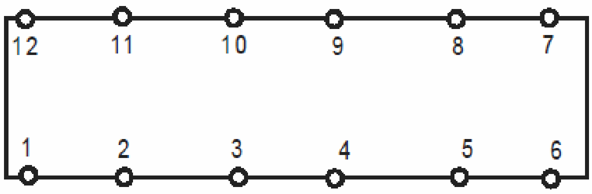
Directions for questions 49 - 50: Answer the questions on the basis of the following information.

In the annual board meeting of Wellcare Inds. Pvt. Ltd., Ashish, Abhiraj, Farooq, Hemant, Jai, Mayur, Mayank, Shahid, Sarwal, Shikhu, Tanya and Vikram are sitting at a rectangular table. The table has 12 chairs numbered 1 to 12 (see the figure). Each chair is occupied by one of the twelve persons (not necessarily in the same order). Some additional information is given below.

1: Farooq, who is sitting opposite Hemant, is sitting at the diagonally opposite end of Mayur, who is sitting on chair number 1.

2: Jai is sitting opposite Sarwal, who is the only person sitting between Abhiraj and Vikram.

3: Ashish is sitting opposite Tanya, who is the only person between Farooq and Shikhu.



49. If Shikhu is not sitting opposite Vikram, then who is sitting next to Mayur?

(1) Abhiraj (2) Jai (3) Vikram (4) Either Jai or Vikram (5) None of these

50. How many different seating arrangements are possible if Mayur is not sitting next to Vikram?

(1) 2 (2) 3 (3) 4 (4) 6 (5) None of these

Directions for questions 51 – 52: Study the following information carefully and answer the questions.

Relationships in family trees are processed by a computer in the form of a string of characters joining letters and symbols. The rules followed by the computer to process the relationships are as below:

X is the sister of Y is processed as X + Y

X is the son of Y is processed as X - Y

X is the mother of Y is processed as X x Y

X is the brother of Y is processed as X divided by Y

X is the daughter of Y is processed as X = Y

X is the father of Y is processed as X not equal to Y

The letters represent a person and the symbol between any two letters defines the relationship that exists between the two persons.

51. If the string of characters that appears as output of the computer for a specific portion of a family tree as its input T + S - R x Q - P, then which of the following relationships exists between P and T?

(1) P is the mother of T. (2) P is the son of T.

(3) R is the daughter of P. (4) T is the uncle of P.

(5) P is the father of T.

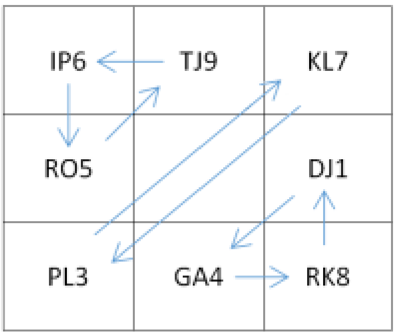
52. If the output generated by the computer is A = B - C × D, then which of the following is definitely false?

(1) B's spouse is the sister-in-law of D. (2) B is the brother of D.

(3) D is the uncle of A. (4) C's spouse is the grandmother of A.

(5) A is the granddaughter of C.

<u>Directions for questions 53 – 55:</u> A 3 × 3 grid changes the contents of its cell by following the instructions in the below steps:



Step-1: For each pair of cells indicated by an arrow, replace the letters of the cell pointed by the arrow head by the ones at the tail of the arrow.

Step-2:

* If the cell contains at least one vowel and an odd number, then change the letters to their second successor in the English alphabet and subtract 2 from the number.

* If the cell contains no vowel, but an even number, then change the letters to their immediate respective predecessors in the English alphabet and add 1 to the number.

Step – 3: Repeat step 1, add 2 to every odd number in the cells, subtract 1 from every even number in the cells.

For a 3 × 3 grid given below, execute the instructions in the three steps and answer the questions that follow.

TP5	KO3	MS4
GI7		JU2
HQ6	RP9	EF8

53. What is the element in the third row of third column in step-2?

(1) SQ9 (2) QO9 (3) LW7 (4) EF2 (5) None of these

54. Which of the following elements is in the third row of second column in step-3?

(1) EF2 (2) EF9 (3) LW11 (4) QO1 (5) FG7

55. Which of the following elements are not the output of the given input in step -3?

(1) WJ5 (2) WL7 (3) TW7 (4) QS9 (5) All of these

Directions for questions:

Segment 1

The decimal number system has a base of 10, i.e. 10 unique digits (0-9) are used to represent all numbers in decimal form. Similarly, a binary number system has a base of 2, i.e. only 2 unique digits (0,1) are used to represent all binary numbers. Decimal numbers 0,1,2,3,4,5 can be represented in Binary as 0,1,10,11,100,101.

To convert decimal to binary, divide the decimal number by 2 repeatedly, recording remainders each time, until the quotient becomes 0 and read the remainders from bottom to top for the binary representation. To convert binary to decimal, start from the rightmost digit, assign a positional value of 2^n, starting with n=0 and incrementing n by 1 for each digit on the left; then, multiply each digit by its positional value and sum for the decimal equivalent. For eg. 21 in decimal is equivalent to 10101 in Binary.

Segment 2

Addition is an arithmetic operation in which two numbers are taken as input, addition operation is performed on these two numbers and the result is a third number.

For e.g. 2+3 = 5. Here 2 & 3 are input numbers, "+" is the addition symbol and 5 is the output number.

Similarly, imagine a hypothetical function called Xation, which uses only two numbers namely 0 & 1 for input & output. Xation function works as shown in the adjacent figure.

Input number 1	Input number 2	Xation	Output number
0	0	X	1
0	1		0
1	0		0
1	1		1

Segment 3

In prefix notation, operators come before numbers. For example, instead of **6 + 8** you write **+ 6 8** and **12 * (9 - 3)** becomes *** 12 - 9 3**.

56. The output of following prefix expression (- + 7 * 45 2 2) and 1011111 are fed into xation function as input. What will be the output?

(1) 1011111 (2) 1111111 (3) 0000000 (4) 0100000 (5) None of these

Directions for questions:

SpaceX is planning their next rover mission to Mars. Below is the miniature map of the landing site. You have to plan the route for the rover from "Start" to "End". Rover can move freely in all blank squares and except squares marked as volcano, crater, lake or mountain. Squares marked as X, Y & Z are unknown as of now and can be assumed as blank squares unless specified otherwise. Rover is initially at square "Start", facing the square marked as X. Coordinates of square Y are described as (3,6). Rover understands only these 4 instructions: Move forward, Turn right, Turn left and Repeat (f,n). Here f can be any of the other three instructions and f is repeated n times. Repeat instruction is coded as two instructions while the other three instructions are coded as single instruction.

Example of route plan with 5 instructions assuming X, Y & Z are blank:

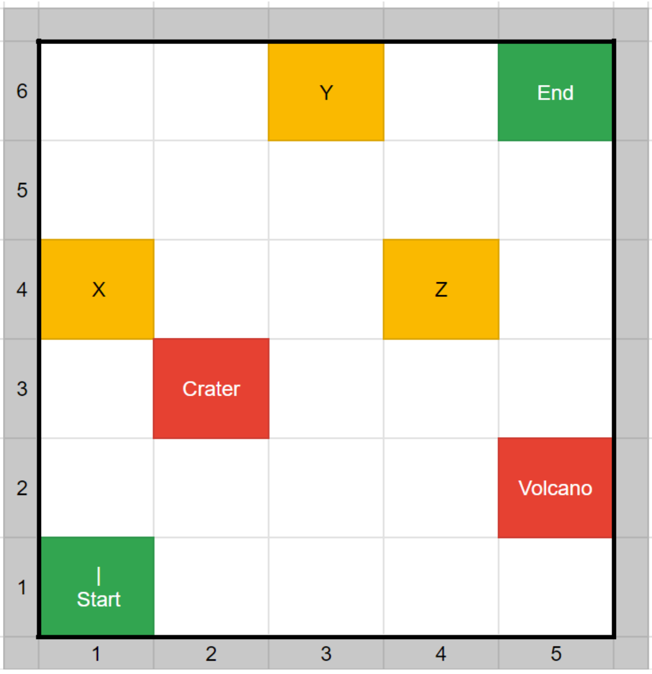
Start

Repeat (Move forward, 5)

Turn right

Repeat (Move forward, 4)

End



57. While the rover is landing, the early spatial scans reveal that square marked as X is an Ammonia lake, square Y is snow mountain and square Z is storm prone area. As Mars rover is ~300 Million KMs far away from Earth station, each instruction that is sent from Ground Station on Earth to the rover on Mars takes around 15 minutes. What is the minimum time (in hours) in which this communication can be completed?

(1) 1 (2) 2 (3) 3 (4) Cannot be determined (5) None of these

<u>Directions for questions 58 – 60:</u> Study the following information carefully and answer the questions that follow.

A machine is given an input in form of a string of numbers.

The machine then executes sequentially the instructions stated in the steps below.

The result obtained after execution of the instruction in any step serves as the input for the subsequent step.

Step 1: Square each number of the input and add 1 to every result.

Step 2: Alternately add 3 and subtract 3 from each number.

Step 3: Multiply each number by 2 and add 2 to it.

Step 4: Subtract the output of step 1 from each number.

Step 5: Add the output of step 3 to each number.

Step 6: Create a new sequence by adding the first and second numbers, then the second and third numbers,

and so on, subtracting 1 from each sum.

58. Let the following string be the first input to the machine.

Input: 3 3 4 5 6 2 8

What will be output of the last step for the given input?

(1) 67 88 136 196 133 217 (2) 67 88 136 133 196 217

(3) 46 22 67 70 127 7 211 (4) 67 88 136 196 133 211

(5) 67 88 136 196 133 217 211

59. Let the following string be the first input to the machine.

Input: 3 3 4 5 6 2 8

What is the sum of all numbers in the output string obtained immediately after execution of the instruction in Step 4 for the given input?

(1) 150 (2) 180 (3) 185 (4) 190 (5) 200

60. If the output of Step 3 is "172 96 138 30 82 0 28", what will be the original input fed to the machine?

(1) 9 7 8 4 6 1 3 (2) 9 7 8 4 5 3 1 (3) 9 7 8 4 5 1 3

(4) 9 7 8 5 4 3 1 (5) 9 7 8 5 4 1 3

SECTION – 4: ENGLISH

Directions for questions 61 – 64: Read the passage and answer the following questions.

Historiography is the study of how history develops truth and knowledge over time. Structures of truth and knowledge are defined as historical methodology. A historiographer can identify the philosophical position of written historical material and determine what epistemology (theory of knowledge with regard to its methods, validity, and scope, and the distinction between justified belief and opinion) is built into the narrative position. Though each structure develops its own epistemology, without a shared notion of collective human experience, there would be no collective history to reflect upon.

Wilhelm Friedrich Hegel introduces a foundational structure for approaching history with "Original History"; a "re-presentation" of the events and experiences that an individual witnesses. It is a record of what has happened, similar to a journal entry. This does not include the ballads or legends of a culture, which is a separate category. The original historian strives to record the exact events witnessed with "life like descriptions".

Hegel also introduces Reflective History, which "transcends the present". The historian looks at the collective history of a people that is recorded. Reflective history is not an exact representation of what happened. It is imbued with the power of the individual's reflective thought. The past can be shaped through the individual's present narrative and becomes "didactic".

The history of ideas is the system of reflective history that studies the Arts, Law, and Religion. These are not observable events, but creations or manifestations of ideas. The Idea is the universal substance that history flows through; this is what connects people to a Universal History. The will of the Spirit generates this activity, not the will of man; because the Idea, the substance of thinking, comes from and is made by the Spirit, not the material activities of man.

Marx states that the conditions for historical study should be verified by experience, and this makes it true. Ideas are not real. They do not exist in the world, and cannot be proven with evidence, therefore irrelevant to study. History is a relational observation of the empirical observable world. Man is therefore defined by how he relates to his natural circumstances.

One notable attempt to bridge Hegel's Idealism with Marx's Materialism, was by Wilhelm Dilthey. He poetically preserved and reawakened the tradition of philosophical idealism that Marx so passionately worked to dismantle.

Wilhelm Dilthey presents an "historical being", which is constructed by the historical process. Human nature is shaped over time by the interactions of language, place and events. The present moment is a culmination of these interactions. Individual inquiry is affected by relative time and place. Their position is a causality of the conditions that came before. Therefore the individual is "historical." The conditions and framework that shape the present moment derive from the "connections in the mind affected world". The mind generates and produces the framework to view the world, and then observes the framework that has been created. This line of reasoning is Dilthey's foundation for knowledge, which has its origins in Hegel's Reflective History.

Whether Marx, Hegel, or Dilthey were "right" or "wrong" is inconsequential. Each individual, through the process of historical contemplation, have outlived their own time and affected the outcome and methodology of history for over two centuries.

61. A historiographer may determine all of the following by analysing an ancient written text, EXCEPT:
    (1) The criteria used to establish the reliability of information given in the text
    (2) The parameters for the qualitative analysis of a creative piece of that time
    (3) The extent of topics, subjects, or themes covered in the written work
    (4) The distinction between fairly credible and speculative interpretations
    (5) None of these

62. All of the following constitute a point of difference between the "Original History" and the "Reflective History" EXCEPT that:
    (1) Original History is concerned with an objective view of events, while Reflective History is concerned with a subjective view.
    (2) Original History includes cultural lores dedicated to historic figures, while Reflective History does not.
    (3) Reflective History may include exaggerated versions of events, while Original History does not.

(4) Reflective History is meant to be instructional in nature, while that may not be the case for Original History.

(5) None of these

63. With which of the following statements about the study of history would the followers of Marx agree?

(1) The conditions for historical study are validated by ideas.

(2) Only that which may be observed is true and that which may not be is not.

(3) The history of ideas is conceived by a union of both the human and the divine.

(4) An idea with a physical manifestation may hold the key to understanding history.

(5) None of these

64. On the basis of the passage, which one of the following views can be inferred to be closest to that of Dilthey?

(1) Individuals with similar mental frameworks have similar perceptions of the same event.

(2) An event is interpreted by the mind independent of the conditions that caused the event.

(3) The framework that the mind creates based on the past is subject to observation in the present.

(4) An individual's present position is the culmination of the conditions of causality that he anticipates.

(5) None of these

Directions for questions 65 – 67: Read the passage and answer the following questions.

THE DECADES-OLD proposal to link all of India's major rivers with one another was revived with much fanfare last year. Most political parties welcomed it then as a solution to the country's drinking water and irrigation problems. But it has not taken long for the proposal to come face to face with the hard reality of planning what will be the largest project ever taken up in India. A number of States, from Punjab in the north to Kerala in the south, have expressed their opposition to a transfer of river waters from their territory to other States. The latest example is the considerable anxiety in Kerala about including a link between the Pampa and the Achankovil (flowing through Kerala) and the Vaippar (in Tamil Nadu) in the proposed national river grid. This is only one of many reasons why the ambitious, many would say unrealistic. Schedules for execution of the project have already been thrown out of gear.

The high-level task force on the project, constituted in December 2002, was expected to prepare the schedule for completion of feasibility studies and estimate the cost of the project by the end of April this year. It was to then come up in June with the options for funding the project. It was also expected to convene a meeting in May/June of State Chief Ministers and obtain their agreement and cooperation. None of these deadlines has been met and there is no indication that these events will take place in the near future. This is not surprising, for while the interlinking proposal has been spoken about for decades, all the complex engineering, economic, environmental and social issues involved in the project have never been carefully studied. It is, therefore, not an easy task to draw up in a few months even the time lines for implementation. It will also be impossible to complete within a decade (as decreed by the Supreme Court) execution of a project that at first approximation is estimated to cost Rs. 5,60,000 crores, which is twice the size of India's gross domestic product at present. In fact, the one Government committee that did examine aspects of the proposal to some extent, the National Commission for an Integrated Water Resources Development Plan, was in 1999 ambivalent about the benefits of interlinking the country's rivers.

The drought of 2002 was the context in which the proposal to build a grid connecting India's rivers was revived. Before another drought leads to another round of active interest in the project, it is necessary to come up with answers to two broad sets of questions. The first question is, what will be the total costs and benefits of a river grid project in economic, environmental and social terms. The second will be, what are the different options to meet the future requirements of water and is the interlinking proposal the best among them. Answers to these questions will have to address issues in agricultural technology, patterns of water use, extraction of ground and surface water resources, efficiency in consumption of water in crop cultivation, resource mobilization, human displacement and changes in the environment. A plan on such a scale and of such complexity as the proposal to link the country's rivers can be taken up only after a range of such substantive issues are analyzed threadbare.

65. What is the main idea of the passage?
    (1) To highlight the objections raised by some states against the proposed project to interlink all major rivers of India
    (2) To point out reasons due to which the proposed interlinking of rivers could not be achieved in time
    (3) To critically analyse the pros and cons of such a project
    (4) The political wrangling over a public utility enterprise
    (5) The cost-benefit analysis of a gigantic project

66. Which among the following can be inferred from the passage?
    (1) The most important reason for the delay in the project is the objection raised by some states against it.
    (2) Though the project had been well planned, the delay occurred because of unforeseen circumstances.
    (3) The government Committee set up in 1999 to examine the project, was very positive about the project's outcome.
    (4) The drought of 2002 gave an impetus to the ongoing proposal of interlinking of India`s major rivers.
    (5) The river-linking project could not progress beyond the draft stage because the techno- commercial validation was unconvincing.

67. What is the tone of the passage?
    (1) Sceptical (2) Analytical (3) Pessimistic (4) Hopeful (5) Condescending

Directions for questions 68 – 70: Read the passage and answer the following questions.

Often intelligent design is referred to as pseudo-science; a futile attempt by theists to assert that God exists through the study of science and the reality that human beings occupy. However, when examining the complexities found in the human body it seems impossible to not notice symmetry within the functions of the human body.

When utilising the uniformity of the universe and its laws as a vehicle for pro intelligent design arguments, one must also acknowledge how the human body can serve the same function. All of the separate parts of the human body work in tandem in order to produce a being that has cognitive capabilities that are unmatched by any other species on planet Earth. The human nervous system works in tandem with the brain to allow humans to feel the spectrum of human emotions. Everything from excruciating pain to utter elation is caused by the brain and its relationship with the nervous system.

Secular scholars will assert that these capabilities came by way of random chance and natural selection. When examined, this statement lacks logical validity. For instance, years of innovation and human creation have yielded the computer, an invention capable of processing vast amounts of information. The computer had a designer, and continues to be built upon even today; yet despite all of its capabilities, it pales in comparison to the human brain. Simulating 1 second of human brain activity takes 82,944 processors.

Of all the systems in the human body, one of the most complex is the circulatory system. Evolutionists must explain how the heart and vessels came to be so well laid out within the human body—how the heart came to be protected by a bony cage, and how the vessels are able to navigate around bones. The heart contracts and pumps blood involuntarily; no cognitive thought is required to maintain this necessary act of life and yet the human heart continues to beat.

Secular scholars assert that the symmetry found in nature is just the by-product of the brain's inherent desire to notice symmetry. An omnipotent and omniscient God would create beings with optimal design, and seeing as there are many flaws present in the human body, the idea of a design does not hold ground. This argument, called the dysteleological argument, asserts that random fluctuation and natural selection are both more viable explanations for the evolution of the human body.

However, when examined further, some of the claims of the dysteleological argument are unsubstantiated. The argument is entirely contingent upon what humans perceive to be optimal. It assumes that the human definition of optimal and the Creator's definition are the same. There is no evidence to support this claim, and it rests entirely on an epistemic statement. Furthermore, humans are unaware of the Creator's goals and these imperfections may serve a purpose that we are entirely unaware of.

As we begin to better understand biology and anatomy as well as physics and cosmology in the universe, it is nigh impossible to deny the existence of a design. This fiery philosophical inquisitiveness present in every human desperately desires to know where we came from.

68. 'When utilising the uniformity of the universe and its laws as a vehicle for pro intelligent design arguments, one must also acknowledge how the human body can serve the same function.' Here, the writer implies that:

(1) The structure of the universe advocates for the existence of deliberate design.

(2) Both the universe and human body show evidence of divine intervention.

(3) Human body shares similarities with the universe in terms of its constituents.

(4) Human body is a more accessible proof of the divine design than the universe.

(5) None of these

69. With which of the following statements would secular scholars agree?

(1) Artificial intelligence can never replace human cognition.

(2) Every bodily function developed out of a deliberation.

(3) Trial and error brings about improvement in a system.

(4) Innovation sparks through a moment of divine faith.

(5) None of these

70. While discussing the circulatory system of the human body, the writer bases his argument in favour of a divine intervention upon the fact that:

(1) The functions of the human body are beyond the influence of human mind.

(2) There is an unmistakable symmetry in the design of the human body.

(3) The level of complexity of the human body finds no parallel in nature.

(4) The structure of the human body is such that it ensures human survival.

(5) None of these

Directions: In the following question, four statements are provided between an opening statement 1 and a closing statement 6. The four statements are jumbled up and form a coherent paragraph when properly arranged. Select the alternative representing the proper and logical sequencing of these six taken together.

71. 1. In life, as in literature, people have certain struggles.

A. Often short stories and novels contain some sort of inner struggle, in order to make the plot more interesting.

B. The basic type of struggles known to people is Man Vs Man, Man Vs Nature and Man Vs Himself, otherwise known as inner struggle.

C. In the novels and short stories we read, there are several examples of inner struggles.

D. In real life, inner struggles happen frequently from the littlest things such as thinking how to get some money in order to get the certain things you want that always keep your mind confused or frustrated.

6. There are times when you have within yourself, problems, concerns or questions that you must address.

(1) DCBA (2) ABCD (3) BDCA (4) CABD (5) DABC

Directions: The passage given below is followed by four alternative summaries. Choose the option that best captures the essence of the passage.

72. Regions are conceptually regarded as being under the level of country but above the level of local or municipal bodies. The idea of the region is increasingly being recognised as a globally important component for industrial and economic development. Additionally, regions are becoming the places where processes and patterns of innovation take place, as well as competitiveness in globalisation. Around the world, many regions are attempting to promote new and competitive industries through cross-sectoral inter-organisational collaborations. This is because inter-organisational collaborations are thought to be the key to creating new businesses and industrial promotions in local regions.

(1) The importance of regions has surpassed that of nations because of such regions' role in promoting economic development through collaborations.

(2) There is a greater focus on regions as the drivers of development because of greater innovation and attempts by regions to engage in inter-organisational and sectoral collaborations.

(3) Inherent advantages of regions have led to greater activity and focus towards these small areas as the main contributors towards economic growth and agents of economic collaborations.

(4) Cross-sectoral collaborations have been one of the main drivers of economic growth in regions, transforming these regions into the locus of all economic activity.

(5) None of these

Directions: Read the text and answer the following question.

To mimic sharks' impressive hydrodynamic prowess, ________________ shark-inspired surfaces for the hulls of boats, wind turbines and even high-end swimsuits, all in an attempt to maximize their efficiency in the water. A fine example is the formation of acoustic materials using shark skin as an inspiration. So-called "acoustic materials" are specifically designed to control and manipulate sound waves, usually for the purpose of dampening or transmitting sound. But such a device can only perform the function for which it was created, such as dampening outgoing sound in a submarine.

73. Which choice completes the text so that it conforms to the conventions of Standard English?

(1) the design of material-scientists (2) it has been designed by material-scientists

(3) the materials used by scientists to design (4) material-scientists have designed

(5) None of these

Directions: There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4), the following sentence would best fit.

74. **Sentence:** The renewed public faith in utopian design coincides with this.

Paragraph: Utopianism is having a moment. (1) ________. Everything from the box office success of big-budget science fiction films like Interstellar and The Martian signals our revival of the utopian imaginary. The groundswell of support for the Green New Deal also signifies our desire to construct and inhabit an idealized world. (2) ________. Already, we can see this impulse reflected in the renderings of the UN-endorsed Oceanix project; the Rebuild by Design competition in New York after Hurricane Sandy; and Bjarke Ingels's recently released plan for human settlements on Mars. (3) ________. We know that climate change is an inexorable, existential threat to everything we know and care about on this planet. (4) ________. The urge to escape our ruined planet and start over has never been greater.

(1) Option 1 (2) Option 2 (3) Option 3 (4) Option 4
(5) None of these

Directions: The question has a pair of CAPITALISED words followed by five pairs of words. Select the related pair that does not exhibit the relationship as the question pair.

75. **BLISTERING : GELID**

(1) Seething : Smouldering (2) Unruffled : Frantic (3) Unsullied : Tarnished
(4) Histrionic : Unaffected (5) Turbid : Lucid

Directions: Read the text and answer the following question.

All students who earned advance management degree from Abacus University have received several excellent job offers. Typically, many graduates of Payola College have gone on to pursue advance management degrees at Abacus University. Thus, enrolling as an undergraduate student at Payola College is a wise choice for students who wish to achieve success in their careers.

76. Which of the following, if true, would most strengthen the conclusion?

(1) All students of Payola College have had a choice of career.
(2) Payola College graduates meet the stringent academic standards required for admission to Abacus University.
(3) Several alumni of Payola College do not accept the job offers.
(4) Alumni of Abacus University consider enrolling at Payola College as the first step to a successful career.
(5) Payola College is the only one in the immediate vicinity of Abacus University.

Directions: Read the text and answer the following question.

Many countries are now investing for the long run in new energy technologies. The next milestone in the promise of fusion-powered energy will be the switching-on of the international ITER reactor in the south of France in 2025. But any benefits of fusion are too far off considering the _______ of climate change.

77. Which choice completes the text with the most logical and precise word or phrase?
(1) sustainability (2) significance (3) urgency (4) denial (5) None of these

Directions: In the following question, two statements I and II are given. These statements may be either independent causes, or may be effects of independent causes or of a common cause. One of these statements may be the effect of the other statement. Read both the statements and decide which of the following answer choices correctly depicts the relationship between the two statements.
Mark answer (A) if statement I is the cause and statement II is its effect.
Mark answer (B) if statement II is the cause and statement I is its effect.
Mark answer (C) if both statements I and II are independent causes.
Mark answer (D) if both statements I and II are effects of independent causes.
Mark answer (E) if both the statements I and II are effects of some common cause.

78. I. Many deaths of fish and fauna were reported off the coast of Pacific last week.
II. The Pacific region experienced widespread protests last week from environment enthusiasts.
(1) (A) (2) (B) (3) (C) (4) (D) (5) (E)

Directions: Read the text and answer the following question.
About 14% of magnitude 6.7 or greater strike-slip earthquakes since 2000 have been supershear. That's 50% more than previously thought. Strike-slip earthquakes occur when the edges of two tectonic plates rub sideways against each other. Supershear quakes are a subtype of that group that are caused when faults beneath the surface rupture faster than shear waves—the seismic waves that shake the ground back and forth—can move through rock. As a result, supershear earthquakes ____________

79. Which choice most logically completes the text?
(1) were thought to be rare because scientists had mostly looked for them on land.
(2) tend to cause more shaking, and are potentially more destructive.
(3) are now being extensively researched beyond shores.
(4) slow or block seismic wave propagation and concentrate their energy.
(5) None of these

Directions: The four sentences (labelled 1, 2, 3 and 4) below, when properly sequenced, would yield a coherent paragraph. Decide on the proper sequencing of the order of the sentences and choose the correct option.

80. 1. Political actors engage in a speech act with the potential to transform social relations and recast an issue in terms of survival, urgency, and the need for emergency action.

2. Framing also relates to the construction of threats by way of agenda setting and priming, relating to a wide range of social fields.

3. Although these methodologies have generally been viewed as separate frameworks, both embody similar concepts, terminology, and theoretical developments that suggest elites can portray certain issues as threats and other issues as insignificant.

4. Securitization theory argues that political actors do not respond to security problems; they actively create them by declaring something a 'security' issue.

(1) 1234 (2) 4123 (3) 3124 (4) 2143 (5) 3421

NSAT Phase 2 – Solutions 2

ANSWER KEY

1.	(2)	2.	(4)	3.	(3)	4.	(4)	5.	(2)	6.	(1)	7.	(1)	8.	(3)
9.	(3)	10.	(1)	11.	(4)	12.	(1)	13.	(5)	14.	(3)	15.	(1)	16.	(2)
17.	(3)	18.	(1)	19.	(2)	20.	(1)	21.	(4)	22.	(1)	23.	(2)	24.	(3)
25.	(4)	26.	(3)	27.	(2)	28.	(4)	29.	(2)	30.	(3)	31.	(2)	32.	(5)
33.	(3)	34.	(3)	35.	(3)	36.	(4)	37.	(4)	38.	(5)	39.	(2)	40.	(4)
41.	(1)	42.	(4)	43.	(3)	44.	(4)	45.	(1)	46.	(1)	47.	(4)	48.	(1)
49.	(2)	50.	(3)	51.	(4)	52.	(1)	53.	(5)	54.	(3)	55.	(4)	56.	(2)
57.	(2)	58.	(1)	59.	(4)	60.	(1)	61.	(2)	62.	(2)	63.	(2)	64.	(3)
65.	(2)	66.	(4)	67.	(1)	68.	(2)	69.	(3)	70.	(4)	71.	(4)	72.	(2)
73.	(4)	74.	(4)	75.	(1)	76.	(4)	77.	(3)	78.	(1)	79.	(2)	80.	(2)

Wishing you the best for your upcoming NSAT Exam !